

# Anupam Gupta

## Curriculum Vitae

Department of Physics  
Indian Institute of Technology, Hyderabad  
Telangana, India, 502285  
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### Current Position

- July 2020 - – **Assistant Professor**, *Department of Physics*, Indian Institute of Technology, Hyderabad, Telangana 502285 India.
- July 2020 - Dec 2021 **Assosiate**, *School of Engineering and Applied Sciences*, Harvard University, Cambridge, MA 02138, USA.

### Post-Doctoral experience

- Aug 2018 -June 2020 **Post-doctoral position: Developmental Biophysics**, *School of Engineering and Applied Sciences*, Harvard University, Cambridge, MA 02138, USA.  
Advisor Prof. L. Mahadevan
- Sept 2017 -Aug 2018 **Post-doctoral position: Heat Transfer in multiphase channel flow**, *Department of Mechanical Science and Engineering*, University of Illinois, Urbana, Illinois 61801, USA.  
Advisor Prof. Anthony M. Jacobi
- Sept 2016 -August 2017 **FERMAT post-doctoral position: Migration of complex shape particles in micro-channels**, *Laboratoire de Génie Chimique, ENSIACET, Univ. Toulouse III*, France.  
Advisor Dr. Micheline Abbas
- Aug 2013 -July 2016 **ERC project “DROEMU: Droplets and Emulsions, Dynamics and Rheology”**, *Department of Physics*, University of Rome ‘Tor Vergata’, Italy.  
Advisor Prof. Mauro Sbragaglia

### Education

- 2007-2013 **Ph.D. in Physics**, *Indian Institute of Science*, Bangalore, India. PhD degree awarded on 29 March 2014..
- 2005-2007 **M.Sc. in Physics**, *Jawaharlal Nehru University*, New Delhi, India.
- 2002-2005 **B.Sc. in Physics**, *Deen Dayal Upadhyay Gorakhpur University*, Gorakhpur, India.

### PhD

- Title *Numerical studies of Problems in Turbulence:*  
I Fluid Films with polymer additives;  
II Fluid Films with inertial and elliptical particles;  
III Scaled vorticity moments in three- and two-dimensional turbulence.
- Advisor Professor Rahul Pandit

## MSc

Title    *Grassmann Integration Technique and Random Matrix Theory: Orthogonal Case*  
Advisor    Professor Akhilesh Pandey

## Research Interests

Problems at the interfaces of the non-equilibrium statistical mechanics, fluid turbulence, soft matter, and developmental biophysics. nonlinear dynamics of extended dynamical systems.

- Turbulence in viscoelastic fluid, elastic turbulence, Lagrangian tracking of inertial and elliptical particles in turbulent fluid, two-dimensional turbulence.
- Morphogenesis & Somitogenesis.
- Microorganisms flock in turbulent environment.
- Flows in confined geometries, droplet deformation, breakup, and generation in microchannels.
- Dynamic self assembly of microscale particles/droplets in microchannels.
- Scientific computing of fluid flows in turbulent environment, microchannel, flows with particles.

## Key Words

- Lattice Boltzmann methods, non-Newtonian fluids, thin films, micro- and nanofluidics, multiphase fluid dynamics, emulsions, droplet deformation & breakup.
- Computational fluid dynamics, Navier-Stokes equation, fluid turbulence, Direct Numerical Simulations, pseudo-spectral method, FENE-P model, turbulent transport, force coupling method.

## Research Visits

September, 2013    *Department of Mathematics and Computer Science, Eindhoven University of Technology, The Netherlands.*  
Sept.-Nov. 2012    *NORDITA, Nordic Institute for Theoretical Physics, Stockholm, Sweden.*  
May-June 2012    *The Laboratoire Jean-Alexandre Dieudonné, Université Nice Sophia Antipolis, France.*

## Awards, Prizes, Honours

- DAC funding to attend the TUR programme at Isaac Newton Institute for Mathematical Sciences, Cambridge University (January-June 2022).
- Research Fellowship (Microsoft Research Corporation, India) (July 2012-July 2013).
- Visiting PhD Student Fellowships 2012, NORDITA-Nordic Institute for Theoretical Physics, Stockholm, Sweden. (Sept.-Nov. 2012).
- Senior Research Fellowship of the Council for Scientific and Industrial Research, India (2009 - 2012).
- Junior Research Fellowship of the Council for Scientific and Industrial Research, India (2007 - 2009).
- All India Rank-70 in the Joint Entrance Screening Test (JEST) (2007).
- All India Rank-161 in the Graduate Aptitude Test in Engineering (GATE) (2007).
- UGC Prize for M.Sc. (2005-2007).
- Gold Medal for ranking first in B.SC. in D.D.U. Gorakhpur University (2005).

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## Membership of Professional bodies

- Member of American Physical Society, 2015-2016, 2018-2026.
- Member of The European Society of Rheology, 2014 - Present.
- Member of INFN (Istituto Nazionale Fisica Nucleare, Italy) 2013 - 2016.
- Member of COST Actions MP1305, 2013 - Present.
- Member of Euro Mech, 2011 - Present.

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## Grants

- PI of the grant awarded by NSM-SERC (2020-2023).
- PI of the grant awarded by SERB-MATRICES (2023-2026).
- PI of the grant awarded by SERB-CRG (2024-2027).
- Co-PI to organize HOWTOCFD workshop, grant awarded by NSM-SERC (2023).
- Co-PI to organize Karyashala workshop on Statistical Tools and Modelling in Biophysics, grant awarded by SERB (2023).
- Co-PI to organize meeting on "Statistical Physics and Complex Systems" (MeetStatphysIndia) and a short summer school on "Nonlinear Physics and Statistical Physics", grant awarded by SERB (2023).
- PI of the grant awarded by Global Initiative of Academic Networks, MoE, India (2024).
- PI to organize meeting on "Pattern Formation in Physical and Biological Systems", grant awarded by ANRF (2025).

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## High performance computing grants

- CollTURB: PI of the grant awarded by NSM-SERC (INR 15,00,000/- PARAM SHIVAY, 2021).
- COMPDROP: PI of the grant awarded by ISCR (4,718,590 core hours, FERMI @ Cineca, 2015).
- POLYDROP: PI of the grant awarded by ISCR (7,864,320 core hours, FERMI @ Cineca, 2014).

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## Workshops, Schools and Conferences Organized

- Organized "Pattern Formation in Physical and Biological Systems" Sept 08-09, 2025 at IIT Hyderabad.
- Organized "Coherent Structures in Complex Physical and Biological Systems" Sept 03-07, 2025 at IIT Hyderabad.
- Co-organized "CompFlu-2024" Dec 16-18, 2024 at IIT Hyderabad.
- Co-organized "Pre Conference Workshop: CompFlu-2024" Dec 15, 2024 at IIT Hyderabad.
- Co-organized SERB-Karyashala "Hands-on Workshop on Statistical Tools and Modelling in Biophysics" July 03-13, 2023 at IIT Hyderabad.
- Co-organized "National Supercomputing Mission Workshop for Training in Computational Fluid Dynamics" June 19-24, 2023 at IIT Hyderabad.
- Co-organized SERB conference "Discussion Meeting on Statistical Physics and Complex Systems" June 5-7, 2023 at IIT Kharagpur.
- Co-organized SERB short school on "Nonlinear Physics and Statistical Physics" June 8-10, 2023 at IIT Kharagpur.

- o Organized one day “Hyderabad Soft Matter Day 2023” May 13, 2023 at IIT Hyderabad.
- o Co-organized conference “Discussion Meeting on Statistical Physics and Complex Systems” July 18-20, 2022 at IIT Kharagpur.
- o Co-organized a short summer school on “Nonlinear Physics and Statistical Physics” July 21-23, 2023 at IIT Kharagpur.

## Publications

### Papers Submitted for Publication

1. *Flow-Induced Phase Separation for Active Brownian Particles in Four-Roll-Mill Flow*  
S. D. Prajapati, A Bhatnagar, **A. Gupta**  
arXiv:2510.24960
2. *Motility-Driven Viscoelastic Control of Tissue Morphology in Presomitic Mesoderm*  
S. Islam, M. S. Rizvi, **A. Gupta**  
arXiv:2510.24314
3. *Emergence of local ordering and giant number fluctuations in active turbulence*  
K. Kashyap, K. V. Kiran, **A. Gupta**  
arXiv:2507.04890
4. *Emergent Strategies for Shepherding a Flock*  
A. Rangnathan, D. Guo, A. Hyde, **A. Gupta**, and L. Mahadevan.  
arXiv:2211.04352
5. *A mechanochemical framework for somitogenesis*  
**A. Gupta**, and L. Mahadevan.

### Publications in Peer Reviewed Journals

1. *Effect of Peclet Number on MIPS in four-roll-mill flow*  
S. D. Prajapati, A. Bhatnagar, **A. Gupta**.  
Accepted for publication in De Gruyter proceedings in mathematics
2. *Effect of the background flow on the motility induced phase separation*  
S. D. Prajapati, A. Bhatnagar, **A. Gupta**.  
Soft Matter **21**, 4700 (2025)
3. *Single-molecule tracking reveals the dynamics of Ipl1 recruitment to the kinetochores and spindles in *S. cerevisiae**  
N. K. Podh, A. Das, A. Kumari, K. Garg, R. Yadav, K. Kashyap, S. Islam, **A. Gupta**, G. Mehta.  
Life Science Alliance **8**, 7, e202503290 (2025)
4. *The Onset of Intermittency in Active Turbulence*  
K. V. Kiran, K. Kumar, **A. Gupta**, R. Pandit, S. S. Ray.  
Phys. Rev. Lett. **134**, 088302 (2025).
5. *Extracellular volume expansion drives vertebrate axis elongation*  
A. Michaut, A. Mongera, **A. Gupta**, O. A. Tarazona, M. Serra, G. Kefala, P. Rigoni, J. G. Lee, F. Rivas, A. R. Hall, L. Mahadevan, K. Guevorkian, O. Pourquié  
Current Biology: Cell Press **35**, 4, 843-853 (2025).
6. *Single-Molecule Tracking dataset for histone H3 (hht1) from live and fixed cells of *Schizosaccharomyces pombe**  
A. Kumari, N. K. Podh, S. Sen, K. Kashyap, S. Islam, **A. Gupta**, E. Rajakumara, M. Nambiar, G.

Mehta.

*Nature Scientific Data*, **11**, 1393 (2024)

7. *Preserving large-scale features in simulations of elastic turbulence*  
S. R. Yerasi<sup>1</sup>, J. R. Picardo, **A. Gupta**, and D. Vincenzi.  
*J. Fluid Mech.*, **1000**, A37 (2024).
8. *Tuning porosity of macroporous hydrogels enables rapid rates of stress relaxation and promotes cell expansion and migration*  
B. A. Nerger, K. Kashyap, B. T. Deveney, J. Lou, B. F. Hanan, Q. Liu, A. Khalil, T. Lungjangwa, M. Cheriyan, **A. Gupta**, R. Jaenisch, D. A. Weitz, L. Mahadevan, and D. J. Mooney  
*Proc. Nat. Acad. Sci.*, 121 **45**, e2410806121 (2024).
9. *Mobilization of trapped oil droplet in porous media through viscoelasticity*  
V. Dzanic, C. S. Form, Z. Wang, **A. Gupta**, C. Xie, and E. Sauret.  
*Phys. Fluids.*, **35**, 093108 (2023).
10. *Irreversibility in Bacterial Turbulence: Insights from the Mean-Bacterial-Velocity Model*  
K. V. Kiran, **A. Gupta**, A. K. Verma and R. Pandit.  
*Phys. Rev. F*, **8**, 023102 (2023).
11. *Geometry dependence of viscoelastic instabilities through porous media*  
V. Dzanic, C. Soriano, **A. Gupta**, C. Xie, and E. Sauret  
*Physics of Fluids*, **35**, 023105 (2023).
12. *Matrix viscoelasticity controls spatio-temporal tissue organization*  
A. Elosegui-Artola\*, **A. Gupta\***, A. J. Najibi, B. R. Seo, R. Garry, M. Darnell, W. Gu, Q. Zhou, D. A. Weitz, L. Mahadevan, D. J. Mooney (\* Equal contribution.)  
*Nature Materials*, **22**, 117-127 (2023).
13. *Rectified random cell motility as a mechanism for embryo elongation*  
I. Regev\*, K. Guevorkian\*, **A. Gupta\***, O. Pourqui, and L. Mahadevan. (\* Equal contribution.)  
*Development*, **149(6)**, 199423 (2022).
14. *Flocking of Active Particles in a Turbulent Flow*  
**A. Gupta**, A. Roy, Arnab Saha, and S. S. Ray.  
*Phys. Rev. F (Rapid)*, **5**, 052601 (2020).
15. *Effect of polymer-stress diffusion in the numerical simulation of elastic turbulence*  
**A. Gupta**, D. Vincenzi.  
*J. Fluid Mech.*, **870**, 405-418, (2019).
16. *Conditional stability of particle alignment in finite-Reynolds-number channel flow*  
**A. Gupta**, M. Abbas, P. Magaud and C. Lafforgue.  
*Phys. Rev. F*, **3**, 114302 (2018).
17. *Inertial Spheroids in Homogeneous, Isotropic Turbulence*  
A. Roy, **A. Gupta**, and S. S. Ray.  
*Phys. Rev. E (Rapid)*, **98**, 021101(R), (2018).
18. *Heavy inertial particles in turbulent flows gain energy slowly but lose it rapidly*  
A. Bhatnagar, **A. Gupta**, D. Mitra, and R. Pandit.  
*Phys. Rev. E*, **97**, 033102, (2018).
19. *The role of the BKM theorem in 3D Euler, Navier-Stokes and Cahn-Hilliard-Navier-Stokes analysis*  
J. D. Gibbon, **A. Gupta**, N. Pal, and R. Pandit.  
*Physica D*, **376-377** 60-68 (2018).

20. *Droplet breakup driven by shear thinning solutions in a microfluidic T-Junction*  
E. Chiarello\*, **A. Gupta**\*, G. Mistura, M. Pierno, and M. Sbragaglia.  
*Phys. Rev. Fluids*, **2**, 123602, (2017).(\* Equal contribution.)
21. *An overview of the statistical properties of two-dimensional turbulence in fluids with particles, conducting fluids, fluids with polymer additives, binary-fluid mixtures, and superfluids*  
R. Pandit, D. Banerjee, A. Bhatnagar, M.-E. Brachet, **A. Gupta**, D. Mitra, N. Pal, P. Perlekar, S.S. Ray, V. Shukla, and D. Vincenzi.  
*Phys. Fluids*, **29**, 111112, (2017).
22. *Lyapunov dimension of elastic turbulence*  
E. L. C. Vi M. Plan, **A. Gupta**, D. Vincenzi, and J. D. Gibbon.  
*J. Fluid Mech. (Rapid)*, **822**, R4, (2017).
23. *Elastic-turbulence induced melting of a nonequilibrium vortex crystal in a forced thin fluid film*  
**A. Gupta** and R. Pandit.  
*Phys. Rev. E*, **95**, 033119, (2017).
24. *Deviation-angle and trajectory statistics for inertial particles in turbulence*  
A. Bhatnagar, **A. Gupta**, D. Mitra, P. Perlekar, M. Wilkinson, and R. Pandit.  
*Phys. Rev. E*, **94**, 063112, (2016).
25. *A regularity criterion for solutions of the three-dimensional Cahn-Hilliard-Navier-Stokes equations and associated computations*  
J. D. Gibbon, N. Pal, **A. Gupta**, and R. Pandit  
*Phys. Rev. E*, **94**, 063103, (2016).
26. *Lattice Boltzmann Simulations of Droplet formation in confined Channels with Thermocapillary flows*  
**A. Gupta**, M. Sbragaglia, D. Belardinelli, and K. Sugiyama.  
*Phys. Rev. E* **94**, 063302, (2016).
27. *How long do particles spend in vortical regions in turbulent flows?*  
A. Bhatnagar, **A. Gupta**, D. Mitra, R. Pandit, and P. Perlekar.  
*Phys. Rev. E*, **94**, 053119, (2016).
28. *Binary-Fluid Turbulence: Signatures of Multifractal Droplet Dynamics and Dissipation Reduction*  
N. Pal, P. Perlekar, **A. Gupta**, and R. Pandit.  
*Phys. Rev. E*, **93**, 063115, (2016).
29. *Depletion of Nonlinearity in Magnetohydrodynamic Turbulence: Insights from Analysis and Simulations*  
J. D. Gibbon, **A. Gupta**, G. Krstulovic, R. Pandit, H. Politano, Y. Ponty, A. Pouquet, G. Sahoo, and J. Stawarz.  
*Phys. Rev. E*, **93**, 043104, (2016).
30. *Effects of viscoelasticity on droplet dynamics and break-up in microfluidic T-Junctions: a lattice Boltzmann study*  
**A. Gupta** and M. Sbragaglia.  
*Eur. Phys. J. E*, **39:6**, 1-16, (2016).
31. *A Lattice Boltzmann study of the effects of viscoelasticity on droplet formation in microfluidic cross-junctions*  
**A. Gupta** and M. Sbragaglia.  
*Eur. Phys. J. E*, **39:2**, 1-15, (2016). Selected as the front cover of the journal.
32. *Homogeneous Isotropic Superfluid Turbulence in Two Dimensions: Inverse and Forward Cascades in the Hall-Vinen-Bekharevich-Khalatnikov model*  
V. Shukla, **A. Gupta**, and R. Pandit.  
*Phys. Rev. B*, **92**, 104510, (2015).

33. *Two-dimensional, homogeneous, isotropic fluid turbulence with polymer additives*  
**A. Gupta**, P. Perlekar, and R. Pandit.  
*Phys. Rev. E*, **91**, 033013, (2015). Selected for Kaleidoscope.
34. *Hybrid LBM/Finite Difference simulations of viscoelastic multicomponent flows in confined geometries*  
**A. Gupta**, M. Sbragaglia, and A. Scagliarini.  
*J. Compu. Phys.*, **291**, 177, (2015).
35. *Deformation and break up of viscoelastic droplets in confined shear flow*  
**A. Gupta** and M. Sbragaglia.  
*Phys. Rev. E*, **90**, 023305 (2014).
36. *Tuning Drop Motion by Chemical Patterning of Surfaces*  
S. Varagnolo, V. Schiocchet, D. Ferraro, M. Pierno, G. Mistura, M. Sbragaglia, **A. Gupta** and G. Amati.  
*Langmuir*, **30**, 2401-2409, (2014).
37. *Regimes of nonlinear depletion and regularity in the 3D Navier-Stokes equations*  
J. D. Gibbon, D. A. Donzis, **A. Gupta**, R. M. Kerr, R. Pandit, and D Vincenzi.  
*Nonlinearity*, **27**, 1-19, (2014).
38. *Elliptical Tracers in Two-dimensional, Homogeneous, Isotropic Fluid Turbulence: the Statistics of Alignment, Rotation, and Nematic Order*  
**A. Gupta**, D. Vincenzi, and R. Pandit.  
*Phys. Rev. E (Rapid)*, **89**, 021001(R), (2014).
39. *Vorticity moments in four numerical simulations of the 3d navier-stokes equations*  
D. A. Donzis, J. D. Gibbon, **A. Gupta**, R. M. Kerr, R. Pandit, and D Vincenzi.  
*J. Fluid Mech.*, **732**, 316-331, (2013). (Authors in alphabetical order.)

#### Publications in Peer Reviewed Conference Proceedings

1. *Deformation and Break-up of viscoelastic droplets using Lattice Boltzmann Models*  
**A. Gupta** and M. Sbragaglia.  
*Procedia IUTAM*, **15**, 215-227, (2015).
2. *Viscoelastic Multicomponent Fluids in Confined Flow-Focusing Devices*  
**A. Gupta**, M. Sbragaglia, E. Foard, and F. Bonaccorso.  
*AIP Conference Proceedings* **1648**, 850060, (2016).
3. *Dynamics and Rheology of Droplets in confined geometries: a Lattice Boltzmann study*  
**A. Gupta**, M. Sbragaglia, and R. Scatamacchia.  
*Proceedings of 'Microfluidics 2014', Limerick, 10-12, December 2014.*

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#### Referee/Reviewer

- o Phys. Rev. X, Phys. Rev. Lett., Phys. Rev. E, Phys. Rev. Res., J. Flu. Mech., Phys. Fluids, The European Physical Journal Plus, Computers & Fluids, Journal of Non-Newtonian Fluid Mechanics, International Journal of Heat and Mass Transfer.

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#### Teaching

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|-----------------------|--------------------------------------------------------------------------------------------------------------------------|
| Jan–Apr 2025          | <b>Instructor</b> , Department of Physics, IIT Hyderabad, Hyderabad,, Graduate level.<br>Computation Techniques PH52060. |
| Aug–Nov 2024,<br>2025 | <b>Instructor</b> , Department of Physics, IIT Hyderabad, Hyderabad,, Graduate level.<br>Statistical Mechanics PH5240.   |

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|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Aug–Nov 2023, 2024, 2025 | <b>Instructor</b> , <i>Department of Physics, IIT Hyderabad</i> , Hyderabad,, Graduate level.<br>Computational modeling of biological systems MP6020.                   |
| Aug–Nov 2023             | <b>Instructor</b> , <i>Department of Physics, IIT Hyderabad</i> , Hyderabad,, Graduate level.<br>Mathematical and Computational Methods for Quantum Devices PH5610.     |
| Aug–Nov 2022             | <b>Instructor</b> , <i>Department of Physics, IIT Hyderabad</i> , Hyderabad,, Graduate level.<br>Computational Physics MP5040.                                          |
| Jan–Apr 2022,2023,2024   | <b>Instructor</b> , <i>Department of Physics, IIT Hyderabad</i> , Hyderabad,, Under Graduate level.<br>Fluid Dynamics EP2228.                                           |
| Aug–Nov 2020,2021,2022   | <b>Instructor</b> , <i>Department of Physics, IIT Hyderabad</i> , Hyderabad,, Graduate level.<br>Classical Physics PH7010.                                              |
| Aug–Nov 2020             | <b>Instructor</b> , <i>Department of Physics, IIT Hyderabad</i> , Hyderabad,, Under Graduate level.<br>Classical Physics PH1017.                                        |
| Jan–Apr 2019             | <b>Co Instructor</b> , <i>Applied Mathematics, Harvard University</i> , Cambridge, MA, Graduate level.<br>Patterns AM217.                                               |
| Jan–Apr 2018             | <b>Co Instructor</b> , <i>Mechanical Sciene &amp; Engineering, University of Illinois at Urbana-Champaign</i> , Illinois, Under Graduate level.<br>Senior Design ME470. |
| Aug–Nov 2009             | <b>Teaching Assistant</b> , <i>Department of Physics, IISc</i> , Bangalore, Graduate level.<br>Advanced Statistical Physics (60 hours).                                 |

## Students

### PhD Students

|                    |                                                                                                  |
|--------------------|--------------------------------------------------------------------------------------------------|
| Premkumar Iyer     | August 2024–Ongoing, <i>Collective behaviour of active swimmers</i> .                            |
| Shruti Nikwadi     | August 2024–Ongoing, <i>Effect of viscoelasticity on active microdroplets</i> .                  |
| Ramesh Kavli       | August 2024–Ongoing, <i>Modelling of active swimmers</i> .                                       |
| Kusum Seervi       | Januray 2024–Ongoing, <i>Motility Induced Phase Separation</i> .                                 |
| Chinmaya Pradhan   | August 2023–Ongoing, <i>Modelling of Cancer Metastasis</i> .                                     |
| Kunal Rai*         | Januray 2023–Ongoing, <i>Molecular Dynamics Simulations of water channels</i> .                  |
| Arvind Kumar       | August 2022–Ongoing, <i>Modeling the dynamics of active microswimmers in a confined geometry</i> |
| Sahil Islam        | January 2022–Ongoing, <i>Pattern formation in biological systems</i>                             |
| Kirti Kashyap      | August 2021–Ongoing, <i>Effect of mechanical properties of ECM on cancer metastasis</i>          |
| Anupriya Dutta Roy | August 2020–Ongoing, <i>Chemotactic fibers in turbulent flow</i> .                               |
| Soni D Prajapati   | August 2020–Ongoing, <i>Effect of turbulence on motility induced phase separation</i> .          |

### Undergraduate Students

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|------------------|-------------------------------------------------------------------------------------------------|
| Priyabrata Patra | August 2023–May 2024, <i>Navigation in turbulent environment using Reinforcement learning</i> . |
|------------------|-------------------------------------------------------------------------------------------------|



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| Om Prakash          | August 2023-May 2024, <i>Study of fingering instabilities in tissue invasion using Phase Field Approach.</i>                                                      |
| Durga Charan Behera | August 2023-May 2024, <i>Study of phase transition between Soft network and Hard network of Epithelial Cell in Vertex Model using Machine Learning Algorithm.</i> |
| A.S. Anudeep        | August 2022-May 2023, <i>COLLECTIVE OLFACTORY SEARCH USING REINFORCEMENT LEARNING.</i>                                                                            |
| Nivedita L S        | August 2021-May 2022, <i>Collective Olfactory Search in a Chaotic Environment.</i>                                                                                |
| Kalyana Naik Banoth | August 2020-May 2021, <i>Motility-Induced Phase Separation in a Chaotic Environment.</i>                                                                          |

## Languages

|         |               |                           |
|---------|---------------|---------------------------|
| Hindi   | <b>Native</b> | <i>Speak, Read, Write</i> |
| English | <b>Fluent</b> | <i>Speak, Read, Write</i> |
| Italian | <b>A2</b>     | <i>Speak, Read, Write</i> |
| French  | <b>A0</b>     | <i>Speak, Read, Write</i> |

## Computer skills

|                       |                                        |
|-----------------------|----------------------------------------|
| OS                    | MacOS, Linux, Windows                  |
| Programming Languages | C, C++, FORTRAN 95, Python             |
| Parallel Programming  | MPI, Openmp                            |
| Symbolic Packages     | Mathematica, Matlab, Matplotlib        |
| Graphics Packages     | GnuPlot, Origin, Xmgr, Visit, ParaView |

## Talks and Posters

1. **Differential motility leads to intestinal organoid budding** - NCBS Bangalore - MechBio: Indian Mechanobiology annual meeting - October 15-16, 2025
2. **Matrix Viscoelasticity and Porosity Governs Collective Cell Behavior and Tissue Patterning** - Florence, Italy - STATPHYS29 - July 13-18, 2025
3. **Emergence of local ordering and giant number fluctuations in active turbulence** - CNR, Rome, Italy - Complex Flows and Complex Fluids - July 8-11, 2025
4. **A mechanochemical framework for somitogenesis** - TIFR Hyderabad - HydSoft24 - July 26, 2024
5. **A mechanochemical framework for somitogenesis** - IIT Hyderabad - Applied Mathematics Day - July 25, 2024
6. **Challenges and complexities in simulating elastic turbulence in polymer solutions** - IIT Hyderabad - TOPICAL MEETING ON COMPUTATIONS OF HIGH-SPEED, TURBULENT, AND COMPLEX FLOWS (CHTC) - June 28-29, 2024
7. **A mechanochemical framework for somitogenesis** - ICTS Bangalore - School and Workshop on Soft and Living Matter: from Fundamental Concepts to New Material Design - August 7-25, 2023

8. **A mechanochemical framework for somitogenesis** - ICTS Bangalore - School and Workshop on Soft and Living Matter: from Fundamental Concepts to New Material Design - Augut 7-25, 2023
9. **Matrix viscoelasticity controls spatio-temporal tissue organization** - PHYSICS COLLOQUIUM - Department of Physics, Michigan Technological University, March 30, 2023
10. **Matrix viscoelasticity controls spatio-temporal tissue organization** - ICTS Bangalore - 8th Indian Statistical Physics Community Meeting, February 1-3 2023
11. **Matrix viscoelasticity controls spatio-temporal tissue organization** - IIT Kharagpur - Complex Fluids Symposium 2022, December 19-21 2022
12. **Bacterial Turbulence: A comparison with its fluid-turbulence counterpart** - Isaac Newton Institute for Mathematical Sciences, Cambridge University, United Kingdom - Mathematical aspects of turbulence: where do we stand?, 4 January - 24 June 2022
13. **Optimal Shepherding of Flocks** - IIT Gandhinagar - Complex Fluids Symposium 2021, December 13-15 2021
14. **Optimal Shepherding of Flocks** - IIT Bombay - Soft Matter : Young Investigators e-Meet 2021, October 14-16 2021
15. **Motility and adhesion gradient induced vertebrate body axis elongation and somite formation** - IIT Bombay - Complex Fluids Symposium 2020, December 10-12 2020
16. **Mathematical modeling of biological systems** - National Institute of Technology, Manipur - National Conference on Applied Physics & Mathematics, December 19-20, 2020.
17. **Mechanical regulation of shape deformation by matrix viscoelasticity in breast tissues** - Denevar, Colorado - APS March Meeting 2020, March 2-6, 2020.
18. **Motility and adhesion gradient induced vertebrate body axis elongation and somite formation** - Boston, Massachusetts - APS March Meeting 2019, March 4-8, 2019.
19. **Motility and adhesion gradient induced vertebrate body axis elongation and somite formation** - Cambridge, Massachusetts - Morphogenesis: Geometry and Physics, Harvard University, MA, December 3-5, 2018.
20. **Statistical studies of non-Newtonian effects in turbulent and confined laminar flows** - Urbana, Illinois - Fluids Seminar at MechSe, UIUC, Illinois, Januray 26, 2018.
21. **How long do particles spend in vortical regions in turbulent flows?** - Toulouse, France - GDR Turbulence Meeting, Insitut de Mècanique des Fluides de Toulouse, June 26-28, 2017.
22. **Hydrodynamic self-assembly of neutrally-buoyant particles in a channel flow: effect of flow inertia** - Naples, Italy - Fluids and Structures: Interaction and Modeling, Hotel Royal Continental, Naples, May 22-26, 2017.
23. **Effects of viscoelasticity on droplet dynamics and break-up in microfluidic T-Junction: a Lattice Boltzmann study** - Porto, Portugal - Flowing Matter 2016, January, 2016.
24. **Effects of viscoelasticity on droplet dynamics and break-up in microchannels: a Lattice Boltzmann study** - Boston, Massachusetts, USA - APS DFD Meeting 2015, November, 2015.
25. **Effects of viscoelasticity on droplet dynamics and break-up in microfluidic T-Junction** - University of Twente, The Netherlands - DROPLETS 2015, October, 2015.
26. **Effects of viscoelasticity on droplet dynamics and break-up in microchannels: T-Junction & Cross-Junction** - Palermo, Italy - FisMat2015, October, 2015.
27. **Effects of viscoelasticity on droplet dynamics and break-up in microchannels: a Lattice Boltzmann study** - Indian Institute of Technology Delhi, New Delhi, India, September 2, 2015.

28. **Effects of viscoelasticity on droplet dynamics and break-up in microchannels: a Lattice Boltzmann study** - School of Physical Sciences (SPS), JNU, New Delhi, India, August 7, 2015.
29. **Effect of Viscoelasticity on Droplet-Breakup and Generation in Microchannels: a Lattice Boltzmann study** - Nantes, France - 10<sup>th</sup> Annual European Rheology Conference, April, 2015.
30. **Non-Newtonian effects on Droplet-deformation and breakup in confined geometries: a Lattice Boltzmann study** - Münster, Germany - Nonlinear Dynamics of Structure Formation at Interfaces, March, 2015.
31. **Viscoelastic Multicomponent Fluids in confined Flow-Focusing Devices** - Hotel Rodos Palace, Rhodes, Greece - 12<sup>TH</sup> International Conference of Numerical Analysis and Applied Mathematics, September, 2014.
32. **Non-Newtonian effects on Droplet-deformation and breakup in confined geometries** - Technical University of Denmark, Copenhagen, Denmark - 10<sup>th</sup> European Fluid Mechanics Conference, September, 2014.
33. **Statistical studies of non-Newtonian effects in turbulent and confined laminar flows** - Harish-Chandra Research Institute, Allahabad, India, August 12, 2014.
34. **Non-Newtonian effects on Droplet-deformation and breakup in confined geometries: a Lattice Boltzmann study** - TIFR Centre for Interdisciplinary Sciences, Hyderabad, India, August 7, 2014.
35. **Non-Newtonian effects on Droplet-deformation and breakup in confined geometries: a Lattice Boltzmann study** - Università Sapienza, Rome, Italy - Superhydrophobicity, bubble stability, and heterogeneous nucleation, June, 2014.
36. **Heavy Particles: Geometry and Persistence** - Indian Institute of Science, Bangalore, India - Transport of Particles in Turbulent Flows, October 2013.
37. **Elastic-turbulence-induced melting of a nonequilibrium vortex crystal in a fluid film with polymer additives** - Università Sapienza, Rome, Italy - Large deviations and rare events in physics and biology, September 2013.
38. **Elastic-turbulence-induced melting of a nonequilibrium vortex crystal in a forced thin fluid film** - the ENS Lyon, Lyon, France - ETC14 September 2013.
39. **Elastic-turbulence-induced melting of a nonequilibrium vortex crystal in a fluid film with polymer additives** - Indian Institute of Science, Bangalore, India - Fluid Days July 2013.
40. **Two-dimensional Homogeneous, Isotropic Fluid Turbulence with Polymer Additives** - Warsaw University, Warsaw, Poland - ETC13 September 2011.
41. **Two dimensional turbulence with polymer additives** - Indian Institute of Science, Bangalore, India - PNLD July 2010.
42. **Polymer additives in two-dimensional turbulence** - The Abdus Salam International Centre for Theoretical Physics, Trieste, Italy - TMB July 2009.

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### Conferences, Workshops and Schools attended

- 'Flowing Matter Across the Scales' (March 24-27, 2015) at the National Institute of Roman Studies, Roma, Italy.
- 'Micro & Nanofluidics Fundamentals and Applications' (May 18-21, 2014) at the University of Twente, The Netherlands.
- 'Turbulence – the Historical Perspective' (September 16-17, 2011) at the Warsaw University, Warsaw, Poland.

- 'Instabilities in Shear Flows' Indo-European network on Advanced Instability Methods Winter Workshop, (Jan 11-15, 2010), at JNCASR, Bangalore, India.
- 'International Conference on Turbulence' (Dec 21-23, 2009) at the Indian Institute of Technology, Kanpur, India.
- 'Tools of Theoretical Physics and the Problem of Turbulence' (February 16-20, 2009) at the S N Bose National Centre For Basic Sciences, Kolkata, India.
- 'Centenary Conference' (December 13-16, 2008) at the Indian Institute of Science, Bangalore, India.
- 'DST SERC School on Nonlinear Dynamics' (June 26-July 16, 2008) at the Indian Institute of Science, Bangalore, India.
- 'International Conference on Nonlinear Dynamical Systems and Turbulence' (July 17-22, 2008) at the Indian Institute of Science, Bangalore, India.
- 'Workshop on Integrable Systems' (February 18-29, 2008) at the Indian Institute of Science, Bangalore, India.
- 'Workshop on Turbulence' (January 07-31, 2008) at the Indian Institute of Science, Bangalore, India.

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